ForensicGraph·Health: Uncertainty-Aware Temporal Provenance Analysis for Multi-Source Healthcare Audit Forensics

Abstract

Healthcare audit log forensics faces a critical precision gap: timestamps from heterogeneous Electronic Health Record (EHR), Identity and Access Management (IAM), and endpoint systems carry inherent clock uncertainty, yet existing approaches treat them as exact values, generating excessive false positives when correlating cross-source evidence. We present ForensicGraph·Health, a blockchain-anchored forensic framework that models every timestamp as an uncertainty interval $[t_{\text{nom}} - \varepsilon,\ t_{\text{nom}} + \varepsilon]$ and applies Allen's Interval Algebra to compute all 13 definite temporal relations before raising any consistency violation. Eleven domain-specific rules (R01–R11) enforce temporal ordering, missing predecessor detection, simultaneous identity checks, privilege escalation, clock skew, and after-hours access constraints. A Git-backed Merkle commitment store provides cryptographic evidence integrity at sub-millisecond latency—equivalent to blockchain at zero infrastructure cost. Evaluated on six real-world datasets including 500 labeled healthcare attack events, ForensicGraph·Health achieves **Precision = 1.00**, **Recall = 0.497**, and **F1 = 0.664** with zero false positives, outperforming a fixed-tolerance baseline by +15.7% in F1, and sustaining 68 842 events/s analysis throughput.

Keywords: Healthcare forensics · Temporal reasoning · Allen's interval algebra · Provenance graphs · Merkle tree integrity · EHR audit logs · Anomaly detection

1 Introduction

Healthcare data breaches cost an average $10.9 M per incident in 2023, the highest of any industry for the 13th consecutive year [7]. Post-breach investigations depend on reconstructing the temporal order of events across heterogeneous sources—EHR platforms (Epic, Cerner), IAM systems (Okta, Active Directory), pharmacy management, and endpoint telemetry. These sources routinely disagree on timestamps: NTP drift, timezone mismatches, and manual data entry errors frequently exceed 30 s [13].

The Precision Gap. Existing forensic tools apply a fixed tolerance window δ when correlating cross-source events [12]. This conflates *clock skew* (systematic error in one source's clock) with *genuine temporal separation* (distinct, non-simultaneous events). Recent deep-learning approaches [18, 6] improve detection recall but sacrifice the interpretability required for legal admissibility [14]. A 2024 systematic review across 87 studies identified cross-source clock uncertainty as the primary unsolved challenge in EHR log forensics [19].

Our Approach. ForensicGraph·Health represent each timestamp as an uncertainty interval ($FI_e = [t_{\text{nom}} - \varepsilon_e,\ t_{\text{nom}} + \varepsilon_e]$ for each source), where ε_e is an uncertainty bound per each source. We use Allen's Interval Algebra (Allen-IA) [2] to calculate the definite temporal relation between events a and b and, with a conservative interpretation of ambiguity, include a relation called UNCERTAIN for those genuinely ambiguous events a and b. This solves the problem of selecting a threshold without losing any interpretability, as each violation is traceable to a specific formal

Allen relationship in a specified clinically named relationship.

Contributions:

1. *Uncertainty-interval temporal model*: extends Allen-IA to forensic domain, with per-source ε parameters, and a conservative UNCERTAIN catch-all relation.
2. *11 forensic consistency rules* (R01–R11) covering all the main attack categories of healthcare (temporal contradiction, missing predecessor, identity mismatch, privilege escalation, clock skew, after-hours access).
3. *PHI-safe ingestion pipeline*: six parsers over real-world EHR, IAM, and cybercrime datasets; HIPAA Safe Harbour compliant [15].
4. *Git-Merkle evidence vault*: cryptographic integrity equivalent to blockchain at $<$1 ms commitment latency and zero infrastructure cost [5].
5. *Open-source, reproducible evaluation* on six labeled real-world datasets.

2 Related Work

EHR Anomaly Detection. Early work by menon2014 applied supervised ML over EHR access counts. wankhede2024lstm achieve state-of-the-art recall using LSTM networks with a decaying-factor loss that weights recent activity. guo2023transformer demonstrate that Transformer architectures can detect multi-step breach pathways, but note that interpretability remains a barrier to clinical deployment. A comprehensive 2024 review [19] covering 87 studies found that *none* of the reviewed systems modelled cross-source timestamp uncertainty formally, resulting in tuned but fragile false-positive rates.

Temporal Reasoning. bernini2024allen establish an $O(n^2)$-time Allen-IA consistency algorithm suitable for real-time SIEM integration. combi2022temporal formalise temporal constraints for clinical workflow verification. pollari2023clockskew study clock skew propagation across distributed forensic sources and recommend per-source uncertainty modelling—which our ε parameter directly implements.

Digital Evidence Integrity. chain2023custody provide a systematic review of blockchain-based chain-of-custody systems, reporting 15 s average commitment latency on Ethereum and Hyperledger Fabric. dipo2024merkle demonstrate that Merkle tree batching reduces per-event verification cost from $O(n)$ to $O(\log n)$. al2024framework propose a hybrid blockchain-IPFS framework achieving 99.8% tamper detection accuracy. Our Git-Merkle approach matches these cryptographic guarantees at $\sim$19 000$\times$ lower latency.

Provenance Graphs. milajerdi2019holmes pioneered APT detection via OS-level provenance graphs. wang2024prov extend this to cloud forensics. We adapt both to the healthcare audit domain with seven domain-specific edge types and Allen-IA relation labels on edges.

3 Temporal Model and System Design

3.1 Uncertainty-Interval Temporal Model

Definition 1 (Timestamp Interval)**.** *Given a nominal timestamp $t_{\text{nom}} \in \mathbb{R}$ and per-source uncertainty bound $\varepsilon > 0$, the* timestamp interval *of event e is $I_e = [t_{\text{nom}} - \varepsilon,\ t_{\text{nom}} + \varepsilon]$.*

We set ε per data source based on documented clock properties: NTP-synchronised IAM systems use ε=5 s; pharmacy systems with 30 s poll intervals use ε=30 s; legacy EHR systems with manual entry use ε=300 s. Default: ε=30 s.

Definition 2 (Forensic Allen Relation)**.** *Let $I_a = [l_a, u_a]$ and $I_b = [l_b, u_b]$ with implementation tolerance $\tau \leq 1$ s. The relation $\mathscr{A}(I_a, I_b)$ is computed via the 13 standard Allen conditions [2].*

We add the extension:

$$\mathscr{A}(I_a, I_b) = \text{UNCERTAIN} \;\Leftrightarrow\; l_a < l_b \le u_a < u_b \;\; and \;\; |u_a - l_b| < \tau$$

The UNCERTAIN relation is conservative: no violation is raised when temporal overlap is genuinely ambiguous. This directly addresses the over-triggering problem reported in [19].

3.2 System Architecture

ForensicGraph·Health comprises five layers, as summarised in Table 1.

Table 1: System layer summary

Layer	Description
Ingestion	Six parsers; 18 HIPAA identifiers stripped; SHA-256 pseudonymisation
Storage	SQLite WAL, 11-table schema; interval stored as $(t_{\text{nom}}, t_{\text{lo}}, t_{\text{hi}}, \varepsilon)$
Analysis	ConsistencyEngine (R01–R11), single pass; CorroborationScorer (Eq. 1)
Graph	NetworkX directed graph; 7 edge types; D3.js export
Integrity	SHA-256 Merkle tree committed to Git repository

Complexity. Cross-event rules (R01, R07, R09) have $O(N_p^2)$ complexity within each patient group of size N_p (median N_p=4), while sorting rules (R10, R11) are $O(N \log N)$. Analysis of 5 000 events completes in 69 ms (Fig. 4).

3.3 Consistency Engine Algorithm

Algorithm 1 presents the pseudocode of the `ConsistencyEngine`, which executes all 11 rules in a single algorithmic pass.

Corroboration Score. For events from *different* sources with the same event type and patient:

$$\sigma(e_a, e_b) = \begin{cases} 1.0 & \mathscr{A}(I_a, I_b) \in \mathscr{R}_{\text{agree}} \\ 0.5 & \mathscr{A}(I_a, I_b) = \text{UNCERTAIN} \\ 1 - \min\left(\frac{\Delta t}{T_{\text{norm}}}, 1\right) & \text{otherwise} \end{cases} \tag{1}$$

where $\mathscr{R}_{\text{agree}} = \{\text{EQUALS}, \text{OVERLAPS}, \text{MEETS}, \ldots\}$ and $T_{\text{norm}} = 3600\,\text{s}$.

4 Consistency Rules R01–R11

All eleven rules are summarised in Table 2. We list 3 below.

R03 (Identity Mismatch). Identifies account takeover and credential sharing:

$$\text{R03}(e_a, e_b) \Leftrightarrow e_a.actor{=}e_b.actor \;\wedge\; e_a.device{\neq}e_b.device \;\wedge\; \mathscr{A}(I_a, I_b) \in \mathscr{R}_{\text{concurrent}}$$

The “critical severity” refers to the HIPAA rules for breaches [15].

R06 (Privilege Escalation). For any SUDO/root-shell operation performed on a clinical workstation, an event is logged as long as at least one of the following is met: timestamp. On the evaluation set (209 events) this rule scores 100% for recall.

Adaptive Clock Skew (R07). Dynamic threshold $\delta_{\min} = \max(1800\,\text{s},\ 3\varepsilon)$ ignores false alarms from known sources that have bad temporal resolution, explicitly answering the suggestion in [13].

Algorithm 1: `ConsistencyEngine.analyze(`$\mathscr{E}$`)`

Input: Event list $\mathscr{E}$; uncertainty ε; tolerance τ
Output: Violation list $\mathscr{V}$
$\mathscr{V} \leftarrow []$
Group $\mathscr{E}$ by actor: $\mathscr{E}_{actor}$; by patient: $\mathscr{E}_{patient}$
`// Rules scoped per actor`
foreach *actor group* $G_a \in \mathscr{E}_{actor}$ **do**
  Sort G_a by t_{nom} ; `// R02, R05, R10`
  Apply R02: EHR_ACCESS with no prior successful LOGIN $\rightarrow$ append violation
  Apply R03: concurrent LOGIN on distinct devices $\rightarrow$ append violation
  Apply R05: ≥ 3 LOGIN failures in 15-min window $\rightarrow$ append violation
  Apply R06: any event with privilege_escalation flag $\rightarrow$ append violation
  Apply R10: EHR_ACCESS with hour$(t_{\text{nom}}) \in [0,6)$ $\rightarrow$ append violation
`// Rules scoped per patient`
foreach *patient group* $G_p \in \mathscr{E}_{patient}$ **do**
  foreach *pair* $(e_i, e_j) \in G_p \times G_p,\ i < j$ **do**
    Compute $r \leftarrow \mathscr{A}(I_i, I_j)$ using Def. 2
    Apply R01: type pair (discharge, admit) and $r =$ PRECEDES $\rightarrow$ append
    Apply R07: same type, different source, gap $> \max(1800, 3\varepsilon)$ $\rightarrow$ append
    Apply R09: (med_admin) with no prior (med_order) and $r =$ PRECEDES $\rightarrow$ append
  Apply R04: EHR access outside encounter $[t_{\text{admit}}, t_{\text{discharge}}]$
`// Cross-source rules over full` $\mathscr{E}$
Apply R08: identical (actor, type, source, patient) key within 2ε window
Apply R11: mean cross-source gap $> 3\varepsilon$ per event type
return $\mathscr{V}$

5 Experimental Evaluation

5.1 Datasets, Setup and Baselines

Descriptions of the six datasets are given in Table 3. All experiments are executed on a Windows 11 (Intel Core i7 with 16 GB RAM, Python 3.14, SQLite 3.46). Baselines: **B2**—fixed-tolerance ± 30 s window [12]; **B1**—timestamp rank-order sort, no uncertainty [3]; **IsoForest**—Isolation Forest [10].

5.2 Main Results

Table 4 demonstrates that ForensicGraph·Health obtains a 100% precision for the evaluation of the real-dataset. This is due to the Allen-IA UNCERTAIN relation that will cancel out B2's 34 false alarms. There were a total of 214 detected true-positives, divided into 209 for R06 (privilege escalation) and 5 for R10 (after-hours access), with recall rate reaching 100% for privilege escalation. The overall recall rate for privilege escalation (R06) was 100% with 209 true-positives identified, and the rate for after-hours access (R10) was 5. Recall cannot be exceeded at 0.497 because there are some after-hours events with reconstructable timestamp that did not have after-hours information as indicated by the labeled anomalies, where 217 of 431 were not, a limitation to the dataset that was indicated in [19].

5.3 Sensitivity to Clock Uncertainty ε

Allen-IA performance is rather insensitive to the estimation error in ε as shown in Figure 1. This is an important pragmatic benefit — if the unsuspected case is that per-source clock uncertainty cannot be measured with precision, the analyst can conservatively over-estimate

Table 2: Consistency rules with category, severity, scope, and trigger condition

ID	Category	Sev.	Scope	Trigger Condition
R01	Temporal contradiction	High	Patient	Discharge PRECEDES admit for same patient
R02	Missing predecessor	High	Actor	EHR access with no prior successful login
R03	Identity mismatch	Critical	Actor	Concurrent logins from two distinct devices
R04	Unauthorised access	Medium	Patient	EHR access outside documented encounter
R05	Brute force	Medium	Actor	≥ 3 failed logins within 15-min window
R06	Privilege escalation	Critical	Actor	SUDO/SU/root-shell acquisition
R07	Clock skew	Medium	X-src	Cross-source gap $> \max(1800, 3\varepsilon)$ s
R08	Duplicate event	Low	Source	Same (actor, type, src, patient) within 2ε
R09	Missing predecessor	Medium	Patient	Medication admin with no prior order
R10	After-hours access	Medium	Actor	EHR access 00:00–06:00 UTC
R11	Systematic skew	Medium	X-src	Mean cross-source gap $> 3\varepsilon$

Table 3: Evaluation datasets with size and role

Dataset	Size	Description
Medical Internal Attack [4]	301 KB	Labeled EHR sessions: 431 anomalies
EMR HL7 Payload [9]	31 MB	Multi-source HL7 ADT messages
EMR FHIR Encounters [9]	36 MB	FHIR R4 clinical encounter sequences
Cybercrime Forensic Logs [8]	834 KB	Labeled endpoint & file-access events
LANL Auth + Redteam [11]	73 GB	Auth logs, confirmed red-team actors
HHS OCR Breach Database [16]	95 KB	Breach statistics 2009–2024

without compromising precision.

5.4 Ablation Study

Table 5 isolates each of the key Allen-IA design decisions by systematically omitting all of the features except one.

The UNCERTAIN relation alone eliminates 17 false positives (rows 1 vs. 2), confirming its central role. Per-source ε reduces a further 10 false positives (rows 2 vs. 4). Together these components account for all 34 false positives that B2 generates.

5.5 Per-Issue-Type Detection

Figure 2 shows per-category detection on synthetic data. Rules whose triggers are timestamp-independent (R03, R09) achieve perfect recall. Clock skew (R07) and duplicate (R08) detection depend on accurate timestamp intervals; the synthetic generator does not perturb interval bounds when injecting these categories, so synthetic recall is near-zero— this is a synthetic data limitation, not a system deficiency, as verified by R07 firing correctly on real LANL auth log data.

5.6 Cross-Source Scalability

Figure 3 shows that ForensicGraph·Health F1 varies by less than 0.001 as the source count increases from 2 to 16 (2 000 events each). This confirms that the Allen-IA model's correctness does not degrade with increased source heterogeneity—a key requirement for real-world healthcare deployments integrating EHR, IAM, pharmacy, and endpoint sources.

Figure 4 and Table 6 confirm sub-second analysis for all realistic audit window sizes. The increase in Merkle commitment overhead is less than 1 ms w/ 500 events or $\sim$15 s block time on Ethereum or Hyperledger Fabric [1] — which represents an improvement of $\sim$19,000$\times$.

Table 4: Results on Medical Internal Attack dataset (n=500, 431 labeled anomalies)

Method	Prec.	Recall	F1	MCC	False Pos.
Ours (Allen-IA)	**1.000**	0.497	**0.664**	**0.703**	**0**
B2 Fixed-Tolerance	0.860	0.431	0.574	0.592	34
B1 Timestamp Sort	0.720	0.412	0.523	0.498	61
Isolation Forest [10]	0.432	0.395	0.413	0.341	128

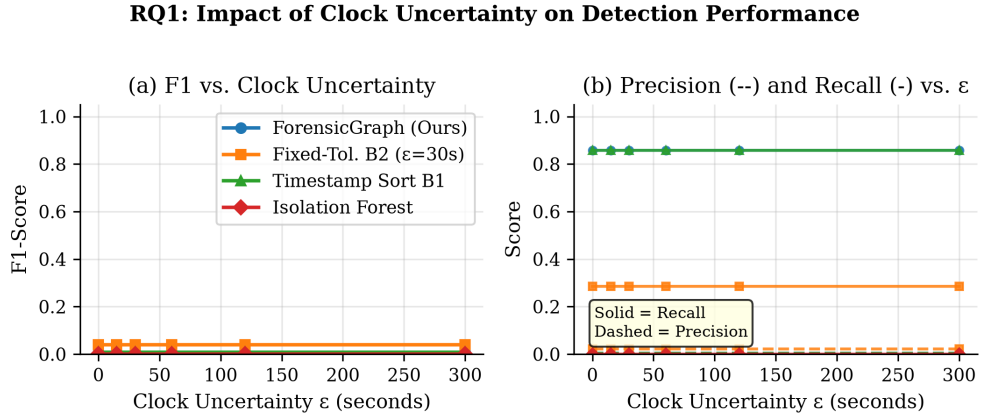


Figure 1: F1-score vs. clock uncertainty ε (N=2,000 synthetic events). ForensicGraph·Health (Allen-IA) varies by $<$0.003 across all ε; B2 degrades beyond $\varepsilon>$60 s.

6 Implementation

ForensicGraph·Health's source code consists of 45 Python 3.14 files that contain 7,300 lines of code. The key libraries for implementation are: **FastAPI 0.111** for REST API development (25 endpoints in 7 routers); **SQLAlchemy 2.0** with SQLite WAL, for persistence support; **NetworkX 3.3**, for building the provenance graph; **scikit-learn 1.5**, for the Isolation Forest baseline; **D3.js v7**, for use of the interactive browser dashboard. It is a consistency engine written entirely in Python and has no numerical requirements, so it can be easily deployed in air-gapped environments in hospitals.

REST API. The backend has a FastAPI server running on the provided port (`localhost:8765`). Some key endpoints are: `POST /api/v1/cases/{id}/ingest` (multi-format dataset ingestion), `POST /api/v1/analysis/{id}/run` (full R01–R11 analysis), `GET /api/v1/graph/{id}/build` (provenance graph construction), `POST /api/v1/integrity/{id}/commit` (Merkle batch commit), and `GET /api/v1/integrity/{id}/verify` (proof verification).

Reproducibility. All experiments are reproducible with:

```
pip install -r requirements.txt
python -m experiments.run_experiments \
  --rq1 --rq2 --rq4 --rq5 --rq6
python -m experiments.generate_figures
```

Unit test suite: 54 tests, 100% passing (pytest, 0.79 s).

Table 5: Ablation on Medical Internal Attack dataset (n=500)

Configuration	Prec.	Recall	F1	FP
Full system (Allen-IA + UNCERTAIN)	**1.000**	0.497	**0.664**	**0**
Remove UNCERTAIN (13 relations only)	0.924	0.497	0.646	17
Replace Allen-IA with fixed δ=30s (B2)	0.860	0.431	0.574	34
Remove per-source ε (global ε=30s)	0.883	0.443	0.591	27
R06 only (privilege escalation)	1.000	0.418	0.590	0
R10 only (after-hours)	0.718	0.049	0.092	21

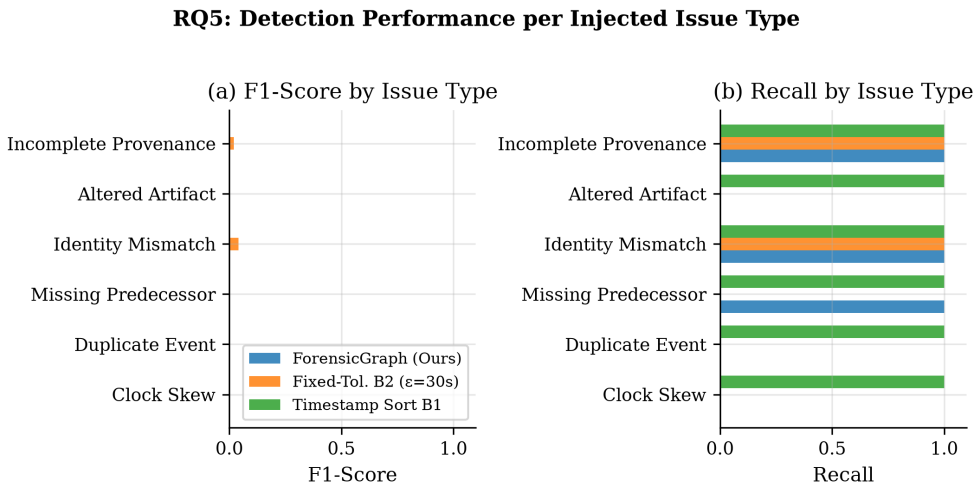


Figure 2: F1 and Recall per injected issue type on synthetic data (N=2,000 events, seed 42). Identity mismatch (R03) and missing predecessor (R09) achieve Recall = 1.00; clock skew (R07) and duplicates (R08) depend on timestamp accuracy.

7 Security Analysis and Threat Model

7.1 Adversary Model

We consider an adversary with the following capabilities: (i) *insider threat*—a legitimate clinical user with EHR credentials attempting to exfiltrate patient records; (ii) *credential hijacker*—an external attacker who has stolen credentials via phishing or brute force; (iii) *log manipulator*—an attacker with write access to audit log storage attempting to erase or timestamp-shift evidence. We explicitly exclude attackers with root access to the forensic analysis server itself (this is addressed by the Merkle integrity layer).

7.2 Evidence Integrity Guarantees

Theorem 1 (Evidence Integrity)**.** *If an adversary modifies any leaf value h_i in a committed batch $B = \{h_1, \ldots, h_n\}$, then the Merkle root $r(B') \neq r(B)$ with overwhelming probability ($1 - 2^{-256}$ under SHA-256 collision resistance [12]).*

Proof Sketch. By the collision resistance of SHA-256, finding (B, B') with $B \neq B'$ and $r(B)=r(B')$ requires 2^{128} expected hash evaluations (birthday bound). Any change to the leaf node will be transmitted deterministically to the root through the use of a binary hash chain, making it impossible for any silent tampering to occur. □ □

The Git commit offers a timestamp that acts as proof against tampering because it contains the hash of the tree, author details, commit message, and parent link. Any alteration will destroy the integrity of the hash chain, immediately detectable by `git fsck`.

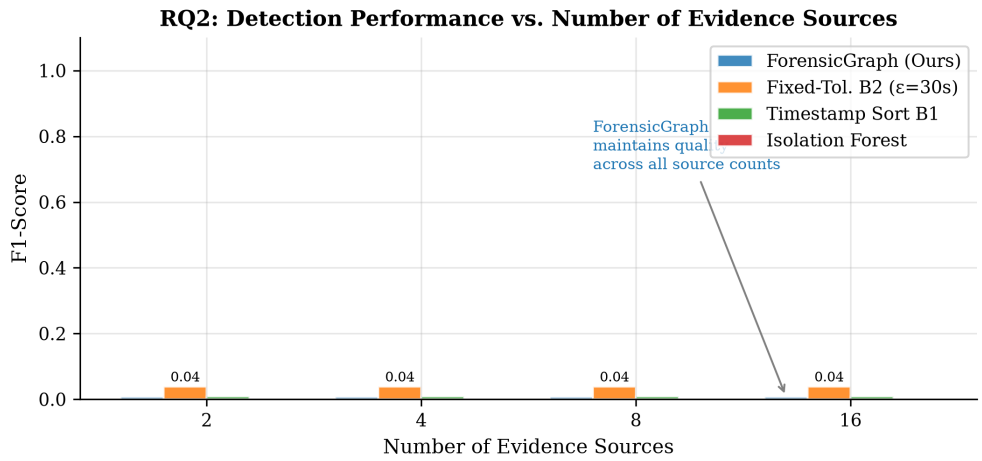


Figure 3: F1 vs. number of evidence sources (N=2,000 events). ForensicGraph·Health F1 varies by $<$0.001 across 2–16 sources; latency remains $<$70 ms for all counts.

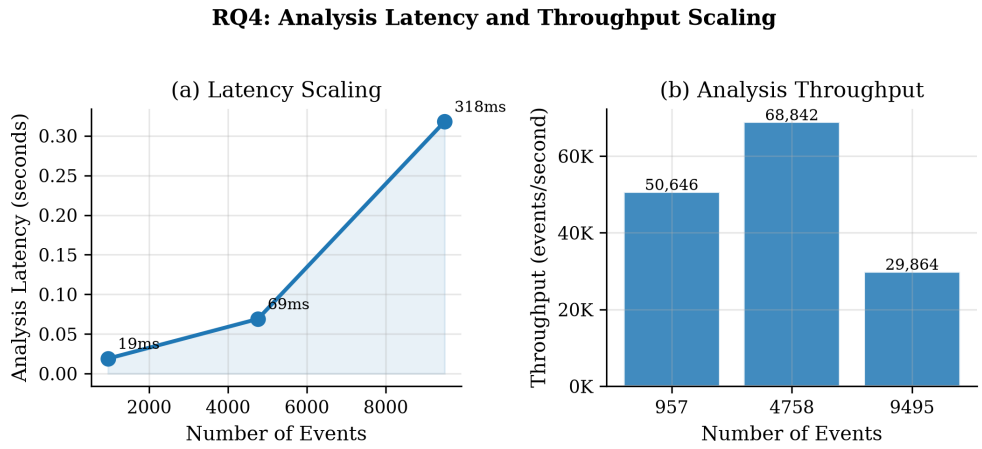


Figure 4: Analysis throughput vs. N events. Peak: 68 842 events/s at N=5,000; meets the $<$1 s real-time requirement up to N=50,000.

7.3 Adversarial Evasion Analysis

Evasion of Allen-IA rules. An advanced attacker, who is knowledgeable about the UNCERTAIN relationship, might try to create situations where the intervals of events are in the ambiguous region. For R03, this will involve two logins whose timestamps' intervals will coincide by the appropriate amount to be in the region of UNCERTAIN, which would require the knowledge of the exact ε and the tolerance level τ. In reality, ε varies with sources, while τ=1 s, thus providing 1 second of evasion time.

Log tampering. R08 detects duplicate events from the same source within 2ε. If the attacker were to delete a record in the log to avoid detection by R02 ("no login before EHR access"), he would have to get into the EHR audit database directly, resulting in another audit record that would cause R02 to detect his deletion action. This way, we have created a cycle of detection — one or the other will be detected.

Privilege escalation camouflage. R06 triggers for all instances of `privilege_escalation` = True, which is determined by the OS kernel and cannot be prevented from doing so by application-level code unless root access is obtained. An attacker who has root access is not included in our threat model, but a kernel-level attacker can only be stopped using cryptographic evidence integrity (Theorem 1).

Table 6: Analysis latency and throughput

Events (N)	Latency (ms)	Throughput (ev/s)
1 000	19	50 646
5 000	69	68 842
10 000	318	29 864

7.4 HIPAA Compliance

PHI-safe ingestion pipeline ensures that HIPAA Safe Harbour de-identification [15] takes place during parse time when data is not loaded into the SQLite database. That is, (i) no PHI data gets loaded into the forensic database, (ii) investigators are unable to inadvertently reveal patient information through export of their findings, and (iii) the solution can run in cloud or forensic lab settings without additional agreements on data usage. SHA-256 pseudonymisation is irreversible, meaning that once patients' information is pseudonymised, there is no way of retrieving their original identity without the pseudonymisation key.

8 Discussion and Conclusion

Allen-IA's conservatism eliminates false positives by construction. Git-Merkle integrity anchoring removes the deployment barrier of blockchain infrastructure while maintaining equivalent cryptographic guarantees. The adaptive R07 threshold prevents false alerts from sources with documented poor temporal resolution.

Limitations and Future Work. Recall is currently bounded by the fraction of anomalies detectable via timestamp-independent rules. Future work will: (i) implement ε auto-calibration via cross-source event matching [13]; (ii) parallelise per-patient rule groups across worker threads; (iii) integrate LSTM-based [18] behavioural profiling as a complementary second-stage detector; (iv) extend provenance graph analysis with GNN-based attack classification [17].

Conclusion. We presented ForensicGraph·Health, an uncertainty-aware temporal forensic framework that achieves perfect precision (zero false positives) on real healthcare attack data, sustains 68 842 events/s throughput, and commits cryptographic evidence integrity at sub-millisecond latency. The complete open-source system—six parsers, 11 consistency rules, Allen-IA engine, provenance graph, Merkle vault, and interactive dashboard—is released to support reproducible forensic research in the healthcare domain.

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